

Index

'Note: Page numbers followed by "f" indicate figures, "t" indicate tables.'

A

- AAS. *See* Atomic absorption spectroscopy (AAS)
- Abrasive particles in CMP slurries, 257–258
- Absorption
 spectroscopy, 626
 spectrum of O₂, 341–342,
 341f–342f
 wavelength, 632
- Accumulation mode particles, 101–102
- Accuracy of system, 633
- Acetone, 15
- Acidic solutions for FEOL post-CMP
 cleaning, 276–278
- Acids, 120–121, 621
- Acoustic streaming, 206–207
- Activated chlorine, 366
- Additives chemistry, 425
- Adsorption, 139–140
 isotherm, 609
 moisture, 125–126
- Aerosol, 461
- AES. *See* Auger electron spectroscopy (AES)
- AFM. *See* Atomic force microscope (AFM)
- Afterglow reactors, 429
- Aging effects, 216–219
- Air
 contamination transport through, 97–98
 exposure, 336
 interaction with hydrogen-terminated
 silicon surfaces, 552–557
 particle measurement in, 691–694, 692t
- Airborne amines, 121
- Airborne molecular contaminants. *See*
 Atmospheric molecular
 contamination (AMC)
- Alarming, trending and, 688–689
- Alcohols reaction with hydrogen-terminated
 silicon surfaces, 566–567
- ALD. *See* Atomic layer deposition (ALD)
- ALE. *See* Atomic layer etching (ALE)
- Al–H₂O system. *See* Aluminum–water
 system (Al–H₂O system)
- Alkaline hydroxylamine solution, 589–590
- Alkoxy groups, 566
- Alkyltrichlorosilanes, 488–490
- “All dry” cleaning, 307–308
- AlN/Al_xO_y, 242
- Aluminum (Al), 8, 254, 434, 580, 581f, 583f,
 625. *See also* Cobalt (Co); Copper (Cu)
 corrosion caused by processing, 587–590
 corrosion pits, 279–280, 280f
 metal gate
 CMP sequence, 266–267, 267f
 post-CMP cleaning solutions, 279–280
 metallization process, 192
 microstructure of Al thin films, 582–583
 organic compound/residue on aluminum
 surface, 279–280, 280f
 passivation, 584–587
 polishing byproducts, 279
 post-CMP cleaning processes, 590
 postetch residue removal, 486
 surface morphology and composition of
 aluminum, 581–584
 wafer defect scan maps, 279–280, 280f
- Aluminum dioxide (Al₂O₃), 9, 257–258,
 279–280, 405, 581–582
- Aluminum interconnect cleaning, 192–195.
 See also Copper interconnect cleaning
 amine-based chemistry, 194
 fluoride-based chemistry, 194–195
- Aluminum–water system (Al–H₂O system),
 279–280
 Pourbaix diagram, 279f
 “AM Clean”, 25
- AMC. *See* Atmospheric molecular
 contamination (AMC)
- Amine-based chemistry, 194
- Amine-based cleaning solutions, 600
- Amine-catechol corrosion inhibitor solution, 589
- Ammonia–peroxide mixture. *See* Standard
 clean 1 (SC-1)—solution

- Ammonium bicarbonate (NH_4HCO_3), 20
- Ammonium chloride (NH_4Cl), 221–222
- Ammonium fluoride (NH_4F), 16, 191, 336
- Ammonium hydroxide (NH_4OH), 17,
187–188, 274, 621
for tungsten post-CMP cleaning, 278–279
- Ammonium hydroxide and hydrogen
peroxide mixture (APM), 187–188,
623
- Ammonium persulfate ($(\text{NH}_4)_2\text{SO}_4$), 16–17
- Amperometric probes, 646
- Analytical measurement technique limits,
674–677
calibration, 674–675
uncertainty and repeatability, 675–677
- Analytical methods, 413
- Anhydrous HF, 329–330
F etching equipment, 363–364
preclean process, 357–358
- Annealing process, 591–592
- Anodic reactions, 594
- APM. *See* Ammonium hydroxide and
hydrogen peroxide mixture (APM)
- Aqueous chemistry of hydrogen-terminated
silicon, 550–551
- Aqueous cleaning and surface conditioning
processes, 186–187
aqueous cleaning equipment, 228–237
BEOL cleaning, 192–197
common chemistries and applications,
197–214
cleaning process sequences, 213–214
metal contamination removal, 211–213
organic contamination removal and
photoresist stripping, 197–201
particle removal, 201–211
current and future challenges, 237–244
back end of line, 241–243
front-end-of-line, 239–241
manufacturing concerns, 243–244
drying, 221–228
applications and techniques, 186–187
FEOL cleaning, 187–192
process variables affecting cleaning,
214–221
rinsing, 186–187, 221–228
- Aqueous cleaning equipment, 228–237
advantages and disadvantages, 234
auxiliary equipment, 235–237
centrifugal spin-spray batch systems, 230
immersion wet benches, 229–230
rinsing processors, 231–234
single-wafer chemical processors,
230–231
- Aqueous HF
aqueous HF solutions, Si (100) and Si (111)
etched in, 528–531
cleaning, 356
- Aqueous media, 169
metallic contamination from, 106–107
- Aqueous-based chemistry methods, 190–191
- Aqueous-based nozzle technology for particle
removal, 209–210, 209t
- Ar/N₂ cryogenic aerosol system,
thermodynamics of, 466–467, 467f
- Arc plasma jet cleaning, 418
- Arc reactors, 438
- Argon (Ar), 421, 426, 583
argon-based cryogenic aerosols, 458
sputter etching, 426
- Arrhenius equation, 478–479
- “Ashing”, 422
- Atmospheric molecular contamination
(AMC), 89, 120–126, 123t–124t,
517–518, 661, 732. *See also* Metallic
contamination; Particle contamination
- acids, 121
bases, 121
condensables, 121–125
dopants, 125
moisture, 125–126, 126f
monitoring, 694
- Atmospheric pressure
CVD, 583–584
plasmas, 438
jet cleaning, 418
torches, 438
- Atomic absorption spectroscopy (AAS),
707t–709t
- Atomic force microscope (AFM), 17–18,
177, 531–532, 736–737, 737f
adhesion force measurement by, 177–178
- Atomic H, 424
- Atomic layer deposition (ALD), 507, 584
- Atomic layer etching (ALE), 380
- Atomic mass units (amu), 722–723
- Atomic mixing zone, 725
- Atomic O, 340–341, 344
- Attenuated total reflectance–Fourier-
transform infrared spectroscopy
(ATR–FTIR), 704, 707t–709t
- Auger electron spectroscopy (AES), 329, 338,
416, 418, 518, 521–522, 703,
707t–709t, 728–730

Auger electrons, 715–716
 Autotitration, 621
 Auxiliary equipment, 235–237
 chemical distribution system, 235–236,
 235f
 particle removal vs. recirculation time,
 238f
 point-of-use filtration, 236–237

B

Back surface particles, 104–106
 Back-end-of-line cleaning (BEOL cleaning),
 8–9, 187, 192–197, 241–243, 264,
 358, 384, 460–461, 473f, 625.
 See also Front-end-of-line cleaning
 (FEOL cleaning)
 aluminum interconnect cleaning, 192–195
 copper dual damascene interconnects,
 242–243
 copper interconnect cleaning, 195–197
 new materials, 241–242
 post-CMP cleaning, 264–272
 two-step room temperature wet cleaning,
 23f
 BARC. *See* Bottom antireflective coating
 (BARC)
 Barrel-type batch reactors, 430
 Bases, 120–121, 621
 Batch
 immersion processing with F-based
 chemistries, 194–195
 processes, 429
 spin-spray systems, 198
 wet processing tool, 638
 wet-chemical cleaning systems, 229
 Beer–Lambert law, 629–630, 630f, 634
 Benzotriazole (BTA), 269–270, 596, 596f,
 599f
 BEOL cleaning. *See* Back-end-of-line
 cleaning (BEOL cleaning)
 BHF. *See* Buffered hydrofluoric acid (BHF)
 Biased-plasma, 447–448
 BiCMOS. *See* Bipolar-complementary MOS
 (BiCMOS)
 Bipolar devices, 390
 Bipolar-complementary MOS (BiCMOS),
 390
 Blanket films, 490
 Blanket test wafers, 668
 BOE. *See* Buffered oxide etch (BOE)
 BOH. *See* Hydroxylated boron (BOH)
 Boltzmann factor, 557–558
 Boron (B), 319–320
 Borophosphosilicate glass (BPSG), 125–126,
 333
 Borosilicate glass vessels, 712
 Bottom antireflective coating (BARC),
 399–400, 423
 BPSG. *See* Borophosphosilicate glass
 (BPSG)
 Bragg scattering, 716
 Brownian motion, 159–160
 Brush break-in, 276, 277f
 Brush scrubbing, 285
 cleaning modules, 287–289
 particle removal mechanism of, 274, 274f
 BTA. *See* Benzotriazole (BTA)
 Bubble cutters. *See* Bubble removers
 Bubble removers, 639–640
 Buffered hydrofluoric acid (BHF), 16, 191,
 522–523, 598, 625
 Buffered oxide etch (BOE), 16, 191
 Bulk etching, 319–331
 etched thickness of BPSG, PSG, and
 thermal SiO₂ vs. partial pressure of
 H₂O, 332f
 etching rate, 325–328
 native SiO₂ regrowth, 329f
 relationship
 of etch time and oxide thickness etched,
 331f
 of HF critical concentration for oxide
 etching and moisture level, 330f
 relative etch selectivity, 331t
 silicon oxide, 328–331
 Bulk photoresist stripping, 33, 386–389,
 387t, 389f
 Bulk silicon defects, 117–118, 118f

C

C₆H₈O₇–based solutions. *See* Citric acid
 –based solutions (C₆H₈O₇–based
 solutions)
 Calibration, 674–675, 687
 of conductivity electrodes, 649
 of electrode measurements, 650
 model, 633–635, 633f, 643
 of pH electrodes, 650
 procedure, 635–638, 637f
 of resistivity electrodes, 650
 results for component A, 643, 652f
 Capillary condensation, 171–173, 172f–174f
 Capillary drying, 27
 Carbon, 399–400, 517–518

- Carbon dioxide (CO_2), 9, 228, 464–465
 - thermodynamics of CO_2 aerosol systems, 464–466
- Carbon-doped oxide (CDO), 137, 406
- Caros acid, 16–17
- Carrier scattering, 119
- Cascade rinsing, 232
- Cassette-less, 229
- Cavitation, 62, 181, 206–207, 684
- CD. *See* Critical dimension (CD)
- CDO. *See* Carbon-doped oxide (CDO)
- Central automated handling system, 363
- Centrifugal spin-spray batch systems, 230
- Centrifugal spray
 - machines, 26
 - processing, 19
 - rinsing, 27
- Ceria (Ce), 257–258
- Cerium oxide (CeO_2), 9, 257–258
 - particles, 264–265, 275–276
- Charge transport, 131–133
- Charge trapping, 356
- Chemical
 - analysis, 634
 - bonding, 171
 - cleaning, 90–91
 - component absorbs light, 626
 - conductivity measurements, 653–654
 - contaminants, 313
 - corrosion, 270
 - distribution system, 235–236, 235f, 689–690, 690f
 - etching, 14–15
 - grown oxides, 509
 - oxidation, 506, 509–510
 - process, 584–585
 - purification methods, 557–558, 713
 - removal, 314–315
 - silicon oxide, 328–329
 - spiking pumps, 218–219
 - suppliers, 235–236
 - vapor deposited Al films, 584
 - vessels, 712
- Chemical mechanical planarization (CMP), 188, 254–264, 295
 - defects, 256f
 - device nodes and defect inspection, 259–260
 - future direction, 293–296
 - performance optimization, 254
 - polishing station tool configuration, 255f
 - for slurry residue removal, 295f
- Chemical mechanical polishing (CMP), 6–7, 195, 385, 460, 513, 590, 663
- Chemical monitoring
 - conductivity and resistivity, 645–646
 - in-line chemical monitoring, 654–655
 - optical absorption spectroscopy, 626–627
 - techniques, 621–623
 - procedure, 626–645
 - wet chemical monitoring, 623–624
 - wet processing chemicals to monitor, 624–626
 - water monitoring and, 620–621
- Chemical oxide removal (COR), 335–336, 353
- Chemical oxides, 199–200, 222, 328–331
- Chemical vapor deposition (CVD), 241–242, 355, 381, 488–490, 582
- Chemisorption, 107–109, 609
- Chemometrics, 628, 631
- Chlorides, 426–427
- Chlorine (Cl), 10, 34, 192, 310, 518, 585
 - atoms, 587
- Chlorofluorocarbon compounds, 15
- Choline, 202
- Citric acid–based solutions ($\text{C}_6\text{H}_8\text{O}_7$ –based solutions), 278–279
- Clean and conditioned wafer surfaces,
 - importance of, 5–9
 - wafer cleaning and surface conditioning for integrated circuit manufacturing, 7–9
 - technology, 6–7
- Clean dry air, 209–210
- Cleaning
 - chemistries, 409–410
 - efficiency. *See* Particle removal efficiency (PRE)
 - particle monitoring on wafer for cleaning tools, 664–668
 - postetch polysilicon gate polymer, 398–403, 402t
 - processes, 380, 506, 581, 589
 - sequences, 213–214
 - steps and equipment requiring particle monitoring, 661–662
 - and surface conditioning, 63
 - technologies, 458
- Cleaning-related defects
 - corrosion, 129–130
 - drying residues, 127–128
 - megasonic damage, 126–127

- oxidation/reduction potentials of metals, 132t
- pattern collapse, 128
- substrate charging, 129
- transient effects during rinsing, 127
- “Clip-on” in-line flow cells, 639
- Closed-loop control system, 623
- Closed-system module, 27
- Cluster tool, 308–309, 308f
 - configuration, 363
- CMOS. *See* Complementary metal oxide semiconductor (CMOS)
- CMP. *See* Chemical mechanical planarization (CMP); Chemical mechanical polishing (CMP)
- CNC. *See* Condensation nucleus counter (CNC)
- Co(II) oxide/hydroxides, 608
- Co(III) oxide, 608
- CO₂-snow (SCO₂), 35
- “Coarse mode” particles, 102
- Cobalt (Co), 4, 270, 580–581. *See also* Copper (Cu)
 - liner, 241–242, 271
 - liner attack, 270–271, 271f
 - passivation process and contamination
 - issues, 607–609
 - post-CMP cleaning, 295–296
 - slit divot defect, 270–271, 271f
 - surface morphology and composition of cobalt, 606–607
- Coincidence event, 718–719
- Commercial tanks, 26
- Compatibility charts, 237
- Complementary metal oxide semiconductor (CMOS), 30, 307
- Complete set of calibration samples, 634
- Complex mixtures, 631–635, 632f
- Composite ceramics (SiO_x/AlO_y), 431–432
- Compound semiconductors, 4
- Concentration
 - concentrated HF, 191
 - distribution, 641–642, 645f
 - effects, 215–216
 - monitoring, 620, 622–623, 655
- Condensables, 120–125
- Condensation, 324
- Condensation nucleus counter (CNC), 690
- Conductivity, 620–622, 645–646
 - chemical conductivity measurements, 653–654
 - measurements, 648–649
 - calibration of electrode measurements, 650
 - installation of electrodes, 650–652
 - effect of temperature on electrode measurements, 649–650
 - ultrapure water resistivity measurements, 652–653
- Conferences on Chemical-Mechanical Polish Planarization for ULSI Multilevel Interconnection*, 7
- Contact angle measurements, 338
- Contact/middle-of-line module, 357–358
- Contaminant(s), 518, 523
 - effects on silicon device, 11–13
 - flux, 96–97
 - types and origins, 10–11
- Contamination, 507–508, 703
 - of Cu passivated surfaces, 597–600, 600f
 - future contamination analysis techniques and trends, 737–738
 - issues with oxidized surfaces, 517–518
 - monitoring programs, 702
 - prevention, 13–14
 - properties in semiconductor environment, 702–703
 - sources during VPD, 712–713
 - transport through air, 97–98
 - transport through liquids, 98
 - of UPW, 651
- Continuous variable filter, 635
- Conventional aqueous methods, 460
- Conventional photodetector, 679
- Conventional PVD metallization techniques, 583–584
- Conventional surface spectroscopic methods, 518
- Conventional thin-film deposition techniques, 602–603
- Copper (Cu), 4, 8, 254, 473–474, 565, 580, 581f, 712. *See also* Cobalt (Co)
 - atoms, 40–44, 41f
 - CMP, 267–268, 267f
 - contamination of Cu passivated surfaces, 597–600
 - Cu-BTAH-H₂O system, 596, 597f
 - Cu/low-k dual damascene processes, 600
 - Cu–water system, potential–pH diagram of, 281f
 - dual damascene interconnects, 242–243
 - films, 426
 - oxide particles, 261–262

- Copper (Cu) (*Continued*)
 passivation processes, 593–597, 594f
 post-CMP cleaning, 267–268
 SEM images of pattern defects and
 comparison of defects counts, 269f
 removal, 494
 surface after cleaning with acidic and
 alkaline solutions, 281–282,
 282f
 zeta potential, 282f
 surface circular scratches, 268, 268f
 surface morphology, 416–417
 and composition, 590–593
 Copper hydroxide (Cu(OH)₂), 594
 Copper interconnect cleaning, 195–197.
 See also Aluminum interconnect
 cleaning
 commercially available fluoride-containing
 BEOL postetch cleaning
 formulations, 196f
 fluoride-based chemistry, 196
 oxidizing-based chemistry,
 196–197
 Copper/low-k dielectric post-CMP cleaning
 solutions, 280–285
 COPs. *See* Crystal oriented pits (COPs)
 COR. *See* Chemical oxide removal (COR)
 Corrosion, 129–130, 584–587
 Al corrosion caused by processing,
 587–590
 dynamics of electrochemical, 130
 galvanic, 270–271, 593
 risk, 258–259
 gross corrosion, 270
 photo-induced, 271–272
 of Co at p and n electrodes, 273f
 micrographs, 273f
 Cosolvents, 496
 Cost, 243
 of ownership model, 243, 430, 640–641,
 654
 Coulomb attraction, 524–526
 Coulombic image force, 469–470
 Critical dimension (CD), 397
 Cryogen, 461
 Cryogenic aerosols, 458–476
 cleaning and conditioning, 35, 60
 surface, 460–461
 fluid cleaning, 34–35
 and formation, 461, 462t
 cryogenic cleaning application examples
 in microelectronic manufacturing,
 472–474
 particle removal mechanism in
 cryogenic aerosol cleaning, 467–471
 thermodynamic properties of gases for,
 462–463, 463t
 thermodynamics of Ar/N₂ cryogenic
 aerosol system, 466–467
 thermodynamics of CO₂ aerosol
 systems, 464–466
 new cleaning technologies, 458–460,
 459f
 processors, 60
 Crystal oriented pits (COPs), 669
 Cunningham correction factor (C_C), 155
 Cupric oxide (CuO), 494, 594
 Cuprous oxide (Cu₂O), 281–282, 494, 594
 Current–voltage characteristics (IV
 characteristics), 335
 CVD. *See* Chemical vapor deposition (CVD)
- ## D
- Damage-free megasonic particle removal,
 182–183
 Damaged polySi line, 207, 208f
 Damascene process, 195, 590–591
 postetch cleaning process, 195
 Darkfield inspection, 670–671, 671f
 Darkfield scattering techniques, 672–673
 “Day” tanks, 235–236
 DC. *See* Direct current (DC)
 DDC. *See* Diluted Dynamic Clean (DDC)
 Decoration effect, 259
 Deep ultraviolet (DUV), 671–672
 Defect(s)
 contamination classification and, 88–92
 according to detection method, 89, 90f
 according to force of adhesion, 91
 according to material, 90–91
 according to size, 91–92
 according to yield and reliability, 92
 and contamination sources, 126–141. *See also* Contamination
 cleaning-related defects, 126–133
 ion implantation, 135
 photoresist, 134
 plasma reactors, 136–138
 process gases, 133–134
 process liquids, 134
 thermal processing, 138–139
 vacuum processing, 139–141, 140t
 wafer handling, transport, and carriers,
 138
 decoration effect, 274–275

- inspection for CMP, 259–260
- size distribution, 95
- Defectivity, 259, 262–264
 - effects on silicon device, 11–13
 - prevention, 13–14
 - types and origins, 10–11
- yield models, reliability, and relationship, 92–96
 - effective area, 95–96
 - statistical yield models, 93–94
- Deformation effect on adhesion, 174–176
- Degree of vibration, 627
- Deionized water (DIW), 8, 186–187, 261, 307–308, 386, 520–521, 588–589, 589f, 625
 - DIW/O₃ solutions, 213–214
 - quality, 219–221
 - rinsing, 356
- Densified fluid cleaning, 476–496
- Density Functional Theory, 553–554
- Depolymerization, 337–338
- Deposited oxides, 335–336
- Deposition process, 582
- Depth of focus (DOF), 663
- Derjaguin, Landau, Verwey, and Overbeek theory (DLVO theory), 167, 257–258
- Derjaguin–Müller–Toporov theory (DMT theory), 174–175
- Design of Experiment (DOE), 642t, 644–645
- Detection method, contamination
 - classification, 89, 90f
- Device nodes for CMP, 259–260
- dHF solution. *See* Dilute HF solution (dHF solution)
- Dialkyldichlorsilanes, 488–490
- Diazo-based materials, 423
- Dielectric(s)
 - barrier films, 437
 - defects in, 119–120
 - films, 330–331
 - metals in dielectric materials, 115–116
 - post-CMP cleaning, 264–266
 - repair applications, 488–494, 489f, 492f
 - hexamethyldisilazane silylation reaction of surface silanols, 489f
 - plasma-damage dielectric thickness effect, 493f
- Differential mobility analysis (DMA), 674–675
- Diffraction methods, 513–514
- Diffusion, 159
- “Digital etch”, 240–241
- Dihydride-dominated spectra, 539
- α -Dihydride”, 549
- Dilute HF solution (dHF solution), 6, 191, 521
- Dilute(d)
 - chemistry, 19
 - citric acid–based cleaning solutions, 279–280
 - HCl, 213
 - NH₄OH for FEOL post-CMP cleaning, 274–276
 - SC-1 solutions, 215
- Diluted Dynamic Clean (DDC), 21, 24
- Diluted sulfuric acid and hydrogen peroxide (DSP), 194, 590, 625
 - DSP+, 194
- Direct current (DC), 684
 - voltage, 722–723
- Dispersive absorption, 638
 - optical absorption, 635
- Dissolved oxygen, 550–551
- DIW. *See* Deionized water (DIW)
- DLVO theory. *See* Derjaguin, Landau, Verwey, and Overbeek theory (DLVO theory)
- DMA. *See* Differential mobility analysis (DMA)
- DMT theory. *See* Derjaguin–Müller–Toporov theory (DMT theory)
- DOE. *See* Design of Experiment (DOE)
- DOF. *See* Depth of focus (DOF)
- Dopants, 120, 125
- Double-layer electrostatic force, 163–166, 163f
- Double-sided PVA roller brush scrubbing, 256
- Downstream
 - generation, 429–430
 - microwave plasma sources, 431–433
 - radio frequency plasma sources, 431
- Drag and lift forces, 154–157
- Drag force, 154–155, 156f
- DRAM. *See* Dynamic random access memory (DRAM)
- Dry cleaning, 468–469
 - and surface conditioning processes, 59–60
 - considerations, 28–29
 - plasma stripping and cleaning, 32–36
 - technologies, 57

Dry cleaning (*Continued*)
 vapor-phase cleaning processes and
 methods, 29–32
 Dry methods of oxide etching, 335–336
 Dry oxide etching process, 336
 Dry-only mode, 233
 Drying, 221–228
 applications and techniques, 186–187
 challenges, 261–262
 defects relating to, 225–228
 liquid between two pattern lines, 228f
 pattern collapse, 227–228
 watermarks, 225–227, 226f
 methods and processors, 223–225
 IPA-based drying, 223–225, 225f
 spin drying, 223
 modules, 292–293
 residues, 127–128
 DSP. *See* Diluted sulfuric acid and hydrogen
 peroxide (DSP)
 Dual damascene
 etching and stripping process flow, 406,
 408f
 processes, 192
 Dummy gate-open CMP, 265, 265f
 Dump rinsing, 231–232
 DUV. *See* Deep ultraviolet (DUV)
 Dynamic random access memory (DRAM),
 241, 458–459, 602
 Dynamic secondary ion mass spectroscopy,
 491–492
 Dynamics of electrochemical corrosion, 130

E

E-beam inspection. *See* Electron-beam
 inspection (E-beam inspection)
 EBR. *See* Edge-bead removal (EBR)
 ECD. *See* Electrochemical deposition (ECD)
 Eckart streaming, 207
 ECR. *See* Electron cyclotron resonance
 (ECR)
 Edge-bead removal (EBR), 674
 EDS. *See* Energy dispersive X-ray analysis
 (EDX)
 EDTA. *See* Ethylenediamine tetraacetic acid
 (EDTA)
 EDX. *See* Energy dispersive X-ray analysis
 (EDX)
 EELS. *See* Electron energy loss spectroscopy
 (EELS)
 EFEMs. *See* Equipment front end modules
 (EFEMs)
 “Effective” electrical defect size, 95
 EIS. *See* Electrochemical impedance
 spectroscopy (EIS)
 Ejector nozzle, 209–210
 Electric field-lines, 159
 Electrical
 damage, 447–448
 double layer, 202
 performance, 331–335
 resistance, 645–646
 Electrochemical deposition (ECD), 590–591
 parameters, 591–592
 Electrochemical impedance spectroscopy
 (EIS), 595
 Electrochemical/electrochemistry, 586–587
 cell, 647
 oxidation of Si, 112
 process, 584–585
 systems, 644
 Electrode(s), 622
 installation, 650–652
 measurements
 calibration, 650
 temperature effect, 649–650
 Electrokinetic adsorption, 221
 Electrolytes, 645
 Electromagnetic spectra, 626–627, 629f
 Electron cyclotron resonance (ECR), 414
 plasma sources, 438
 Electron energy loss spectroscopy (EELS),
 518, 522–523, 531–532
 Electron-beam defect detection systems, 695
 Electron-beam inspection (E-beam
 inspection), 673
 Electronegative elements, 520–521,
 569–570
 Electron–hole pairs, 552
 Electrophoresis, 158–159
 Electrostatic forces, 168–170
 effect of ionic strength, 169–170
 effect of pH, 169, 170f
 Elemental analysis, 702–703
 Ellipsometric parameter, 317, 318f
 Endpoint detection method for stripping and
 cleaning, 440–441
 Energetic radiation, 347
 Energy dispersive X-ray analysis (EDX), 259,
 663–664, 664f, 704, 707t–709t, 728
 Environment, Safety, and Health (ESH),
 200–201, 244
 Epitaxial
 deposition, 413

layers, 8
 Si deposition, 357
 technology, 310
 Equipment front end modules (EFEMs),
 661–662
 ESH. *See* Environment, Safety, and Health
 (ESH)
 Etch residues removal, 33–34
 Etchant gasses, 588–589
 Etching, 319, 494–495
 process, 587, 624, 653
 rate, 325–328
 gas-phase SiO₂, 327f
 predicted SiO₂, 326f
 thermal SiO₂ removed *vs.* time for ratios
 of azeotropic HF/H₂O solutions, 327f
 vapor phase processes, 328t
 variation, 535–536
 of silica, 494–495
 Ethanol, 15
 Ethylenediamine tetraacetic acid (EDTA), 21
 Evaporation, 307, 582
 Excited species, 382–383
 Extrinsic contaminants, 507–508, 569

F

F additives, 403–404
 F-based additives, 425
 F-based chemistry formulation, 194–195
 F-containing
 additive, 392–394
 gas, 394–395
 molecules, 387–388
 plasma, 417
 Fabrication of microelectronics,
 180–181
 Face-centered cubic (FCC), 607
 “Fall-on” mechanism, 105
 Faraday cage, 430
 Faraday shield, 434
 FCC. *See* Face-centered cubic (FCC)
 FEOL cleaning. *See* Front-end-of-line
 cleaning (FEOL cleaning)
 Field distortion effect, 130
 Filters, 236, 666
 Fin field effect transistor (FinFET), 4, 240,
 307, 460–461
 formation CMP, 265, 265f
 transistor, 398–399
 The Fine Particle Society, 7
 FinFET. *See* Fin field effect transistor
 (FinFET)
 Fingerprint region, 627
 Flash memory technology, 241
 Flicker noise, 119
 Fluid flow, 640
 Fluid jet particle removal, 209–210
 Fluorescent X-rays, 718–719
 Fluoride-based chemistry, 194–196
 Fluoride-containing cleaning, 193
 Fluorides, 426–427
 Fluorine (F), 10, 192, 518, 520–521,
 562–563, 569–570
 additives, 425
 contamination, 562–563
 fluorine-based cleaning formulations, 419
 fluorine-containing compounds, 440
 fluorine-containing gases, 399
 passivation of hydrogen-terminated silicon
 surfaces, 567–568
 water reaction with fluorine-terminated
 silicon surfaces, 568
 Fluorocarbons, 403, 425
 resins, 236
 Force of adhesion, classification according to,
 91
 Forced air drying, 27
 Forming gas, 395
 FOUPs. *See* Front opening unified pods
 (FOUPs)
 Fourier transform, 639–641
 Fourier transform infrared spectroscopy
 (FITR), 319, 707t–709t, 733–735
 absorption spectrum of polystyrene, 734f
 of organosilane low-k dielectric film, 284,
 284f
 Friction force, 287
 Front End Processing Surface Preparation
 Technology Requirements, 10t, 12t
 Front opening unified pods (FOUPs),
 661–662
 Front surface particles, 104
 Front-end-of-line cleaning (FEOL cleaning),
 5–6, 8–9, 187–192, 239–241, 264,
 384, 459–460. *See also* Back-end-of-
 line cleaning (BEOL cleaning)
 chemicals, 189t
 four-step room temperature wet-cleaning
 for, 22f
 logic technology, 239–241
 memory technology, 241
 oxide etching and removal, 191–192
 photoresist removal, 190–191
 post-CMP cleaning, 264–272

Front-end-of-line cleaning (FEOL cleaning)

(Continued)

diluted NH_4OH for, 274–276

HF, 274–276

non-SC-1 alkaline solutions, 276–278

SC-1, 274–276

TMAH, 274–276

postetch cleaning, 191

pre-thermal processing cleaning sequences
and chemistries, 190t

preprocessing cleaning, 187–188

RCA chemistries, 189t

resist stripping and postprocess cleaning,
188–191SC-1–SC-2 wet bench cleaning process,
472

Si wafers, 11

FTIR. *See* Fourier transform infrared

spectroscopy (FTIR)

“Functional H_2O ”, 58–59**G**

Gallium arsenide (GaAs), 11

Gallium phosphide (GaP), 11

Gallium-indium alloy, 718

Galvanic corrosion, 270–271, 593

risk, 258–259

GaP. *See* Gallium phosphide (GaP)

Gas-phase cleaning, 309–312

applications, 352–362

Gas-phase H_2O molecule, 319Gas-phase $\text{HF}/\text{H}_2\text{O}$ etching of SiO_2 ,
321

Gas-phase metal removal, 351

Gas-phase wafer cleaning technology, 306,
313f

advantages, 312

applications of gas-phase cleaning,
352–362

best fit of present oxidation model, 355f

chemistry and mechanisms, 313–316, 315f

cluster tool, 308f

contaminants and link with gas-phase
process, 311tcylindrical capacitors after removal of
mold layer, 360fdrivers for gas-phase cleaning and surface
conditioning, 308–310

gas-phase cleaning processes, 310–312

kinetics of organic contamination

deposition, 362f

limitations of wet cleaning, 307–308

 O_3 and UV/O_3 for organic removal, resist
stripping, and surface oxidation,
336–347

process equipment, 362–369

selective hard mask removal, 361f

silicon oxides removal with HF vapor,
316–336 UV/Cl_2 for metallic contamination
removal, 336–347

Gas(es), 421, 421t

chromatography, 633

gas/solid surface reactions, 313–314, 314t

particle deposition mechanisms in, 154

particle measurement in, 691–694, 692t

properties for SCF, 477–478

Gaseous cleaning of Ge, 61–62

Gaseous HF, 357

Gas–solid chemical adsorption reactions,
602–603Gate dielectric breakdown charge (Q_{BD}),
204–205

effects of Si surface roughness on, 205f

Gate dielectrics, 119

Gate electrode, 398–399

Gate module, 354–356

Gate oxidation, 362–363

Gate oxide integrity (GOI), 222

Gate oxide preclean, 362–363

Germanium, 61–62

“Gettering”, 115

Gibbs equation, 462

Gibbs phase rule, 462

Glass transition temperature, 483

“Glow discharge” region, 429–430

GOI. *See* Gate oxide integrity (GOI)

Gold atoms, 40–44, 40f

Grain boundary decoration, 279–280,
280f

Graphic user interface (GUI), 643–644

Gratings, 636

Gravitational force, 158

Group III-V materials, 318–319

Group IV materials, 318–319

GUI. *See* Graphic user interface (GUI)**H**H-bond. *See* Hydrogen bond (H-bond)H-terminated Si surface. *See* Hydrogen-
terminated silicon surfaces
(H-terminated Si surface) H_2 -based plasmas, 405–406 H_2 -based process, 421

- “Half-flat” surface, 542–543
- Halogens, 400–401
- Hansen solubility parameters, 484
- Hard mask trim, 359
- Haze, background scattering, 669–670, 670f
- HDIS. *See* High-dose implants photoresist stripping (HDIS)
- HDP. *See* High-density plasma oxide (HDP)
- Helium (He), 426
- Helms–Deal analysis, 325
- Helms–Deal model equilibrium, 323
- HEPA. *See* High-efficiency particulate air (HEPA)
- Hexamethyldisilazane (HMDS), 423, 488–490, 495
- HF. *See* Hydrofluoric acid (HF)
- HHF approximation. *See* Hogg–Healy–Fuerstenau approximation (HHF approximation)
- HI. *See* Hydroiodic acid (HI)
- High dielectric constant material (High-k material), 8, 57, 130, 239, 266f
 - pre-high-k deposition, 416
- High-density plasma oxide (HDP), 386, 429
- High-dose implants, 199
- High-dose implants photoresist stripping (HDIS), 392–394
- High-efficiency particulate air (HEPA), 26, 89
 - filters, 103–104
- High-k material. *See* High dielectric constant material (High-k material)
- High-mobility materials, 296
- High-performance microprocessor applications, 671–672
- High-resolution IR reflection absorption spectroscopy, 529
- High-atomic weight atoms, 421
- Hillock formation, 544–545
- HMDS. *See* Hexamethyldisilazane (HMDS)
- Hogg–Healy–Fuerstenau approximation (HHF approximation), 165–166
- Hollow cathode plasma arrays, 438
- Horizontal brush scrubber configurations, 287, 287f
- Hot phosphoric acid (H_3PO_4), 218–219, 264
- Hot-wire technology, 438
- Human skin oil, 339–340
- Hydrides, 521–522
- Hydrocarbons, 564–565
 - contamination, 518, 564–565
- Hydrochloric acid (HCl), 187–188, 212–213, 310, 624–625
- Hydrochloric acid hydrogen peroxide mixture (HPM). *See* Hydrogen peroxide (H_2O_2)
- Hydrodynamic boundary layer, 321–322
- Hydrofluoric acid (HF), 188, 191, 202, 213, 264–265, 310, 588–589, 625
 - chemical composition of HF-treated surfaces, 518–523
 - decomposition stage, 710
 - etching, 506
 - for FEOL post-CMP cleaning, 274–276
 - HF-etched Si surfaces, 563–564
 - HF-last processing, 19
 - “HF-last” process, 219
 - HF/ H_2O vapor ratio, 326–328
 - HF/ NH_3 processing equipment, 368–369
 - particle removal, 202–204
 - solution, 9, 15–16
 - vapor etching, cleaning, and surface conditioning, 29–30
- Hydrogen (H_2), 425, 583
 - plasmas, 424
 - termination mechanism, 523–527
- Hydrogen bond (H-bond), 317, 509–512
- Hydrogen ions (H^+), 622
- Hydrogen peroxide (H_2O_2), 6–7, 17, 187–188, 197, 621, 624–625
 - mixtures, 16–17
- Hydrogen peroxide anion (HO_2^-), 197
- Hydrogen-terminated silicon surfaces (H-terminated Si surface), 187, 354, 565–566, 569–570
 - chemical composition of HF-treated surfaces, 518–523
 - contamination and stability
 - aqueous chemistry of hydrogen-terminated silicon, 550–551
 - dissolved oxygen role, 550–551
 - oxidation, 549–550
 - oxygen and air interaction with hydrogen-terminated silicon surfaces, 552–557
 - types, 562–565
 - water vapor interaction, 557–562
- hydrogen termination mechanism, 523–527
- preferential etching mechanism, 545–549
- stability and reactivity, 565–568
 - alcohols reaction, 566–567
 - fluorine passivation, 567–568
 - solvent stability, 565–566

- Hydrogen-terminated silicon surfaces
 (H-terminated Si surface) (*Continued*)
 water reaction with fluorine-terminated
 silicon surfaces, 568
 structure and morphology, 527–545
 Si (100) and Si (111) etched in aqueous
 HF solutions, 528–531
 Si (100) etched in buffered HF solutions,
 537–545
 Si (111) etched in buffered HF solutions,
 531–537
- Hydrogen-terminated surfaces, 514
- Hydrogenated Si surfaces, 99, 100f
- Hydroiodic acid (HI), 310
- Hydrophilic silanol groups (Si–OH), 488,
 489f
- Hydrophilic surface, 222, 283
- Hydrophilic/hydrophobic nature of wafer
 surfaces, 204
- Hydrophobic low-k dielectric film, 286
- Hydroxide ions (OH[−]), 193, 550–551
- Hydroxyl groups (–OH groups), 488–490
- Hydroxyl ions (OH[−]), 622
- Hydroxyl radicals (OH), 605–606
- Hydroxylamine (NH₂OH), 193, 625
 hydroxylamine-based liquid chemical
 cleaning solutions, 358
 hydroxylamine-based residue removal
 mechanism, 194
- Hydroxylated boron (BOH), 318
- I**
- IBD. *See* Ion beam deposition (IBD)
- IC. *See* Integrated circuit (IC)
- ICP. *See* Inductively coupled plasma (ICP)
- ICP–MS. *See* Inductively coupled plasma–
 mass spectroscopy (ICP–MS)
- IEP. *See* Isoelectric point (IEP)
- IHP. *See* Inner Helmholtz plane (IHP)
- III–V epitaxial layers, 4
- ILD. *See* Interlayer dielectrics (ILD)
- IMEC Clean, 21–24, 24f, 213–214
- Imidazole, 596–597
- Immersion
 baths, 229
 systems, 229–230
 tank wet benches, 26
 wet benches, 229–230
- Impurities, 426, 552
- In situ IR spectroscopy, 552, 554–555
- In-line monitoring, 645–653
 chemical monitoring, 654–655
- In-line purifiers, 423
- Index of refraction (RI), 621, 680–682, 681t
- Induction, 318, 647
 period, 318–319
- Inductively coupled plasma (ICP), 387,
 430–431, 433–434
 plasma systems, 434–436, 442
- Inductively coupled plasma sources,
 433–436
- Inductively coupled plasma–mass
 spectroscopy (ICP–MS), 713
- Inert gases, 409, 438
- Infrared (IR), 518, 531–532, 622,
 733–734
 absorption, 521
 radiation, 432
 spectroscopy, 509–510, 520–521, 566
- Inner Helmholtz plane (IHP), 100
- Inorganic analysis, 702–703
- Inorganic contaminants, 343
- Inorganic contamination analysis techniques,
 706–730. *See also* Organic surface
 contamination; Surface contamination
 measurement techniques
 AES, 728–730
 EDS, 728
 SIMS, 720–724
 TOF–SIMS, 724–728,
 724f
 TXRF spectrometry, 715–719, 715f
 VPD, 706–714
 VPD/TXRF, 719–720
 XPS, 730
- Institute of Environmental Sciences, 7
- Instrumental analysis of VPD droplet, 712–713
- Integrated circuit (IC), 4, 154, 380, 579, 620,
 660–661
 cleaning techniques, 509
 devices, 186–187, 458
 manufacturing, 32–33, 306,
 488
 industry, 193
 wafer cleaning and surface conditioning,
 7–9
- Integrated polySi–WSi_x CVD cluster tools, 358
- Integrated processing, 346,
 362–363
- Interconnect post-CMP cleaning, 267–272
- Interferometry, 444
- Interlayer dielectrics (ILD), 488
 films, 264
 deposition, 254

International Organization for
Standardization (ISO), 102–103, 693
ISO 14644–1, 102–103
International Standards Organization. *See*
International Organization for
Standardization (ISO)
*International Symposia on Cleaning
Technology in Semiconductor Device
Manufacturing*, 7
*International Symposia on Ultra-Clean
Processing of Silicon Surfaces*, 7
International Technology Roadmap for
Semiconductors (ITRS), 5–6, 14,
235–236, 458–459, 662
Intrinsic contaminant, 507–508
Ion beam deposition (IBD), 583
Ion-selective electrodes (ISE), 647–648
Ionic/ion
bombardment, 409–410, 412
contaminants, 12
implantation, 135
in solution, 100
strength effect, 169–170
IPA. *See* Isopropyl alcohol (IPA)
IR. *See* Infrared (IR)
ISE. *See* Ion-selective electrodes (ISE)
ISO. *See* International Organization for
Standardization (ISO)
Isoelectric point (IEP), 169, 170t, 202,
257–258
Isolation module cleaning, 353
Isopropyl alcohol (IPA), 15, 30, 121–122,
193, 256, 355–356
IPA-based drying, 223–225, 225f
Isotropic etching processes, 307
ITRS. *See* International Technology
Roadmap for Semiconductors (ITRS)
IV characteristics. *See* Current–voltage
characteristics (IV characteristics)

J

Jet spray, 286
cleaning module, 291–292, 291f
Johnson–Kendall–Roberts theory (JKR
theory), 174–175
Jump-out point, 177–178

K

Karl Fisher reagent for water content, 621
Killer defects, 92–93, 104
density, 93

Kinetic Monte Carlo simulations, 548–549
Kundsen number (Kn), 155

L

Laminar airflow, 97–98
Laminar flow, 98
Langmuir probes, 441
Langmuir regime, 558
Langmuir–Hinshelwood mechanism,
314–315
LANL. *See* Los Alamos National Laboratory
(LANL)
Laplace pressure, 227
Laser
beam, 677
interferometry, 444
irradiation, 419
Latent defects, 92–93
density, 93
Layer-by-layer oxidation, 516–517
LED. *See* Light-emitting diodes (LED)
LEED method. *See* Low-energy electron
diffraction method (LEED method)
Lewis acid–base adduct, 551
Lift-off mechanism of brush scrubbing,
274–275, 275f
Light point defects (LPDs), 668
Light scattering by particle, 669–670
Light wavelength, 442–443
Light-emitting diodes (LED), 635, 644f
Liquid chemicals, 195
particle measurement in, 677–691
NRM, 690–691
OPCs, 677–690
Liquid(s)
cleaning, 29
CO₂, 466
contamination transport through, 98
liquid-phase processes, 331–333
media
measured adhesion forces, 175f,
178–179, 179f
particle deposition mechanisms, 154
particle counters, 677
particle detection, 696
processes
and wafer drying techniques, 14–15
and wafer drying technology,
58–59
Lithium-drifted silicon detector (SiLi
detector), 718–719
Lithographic media, particles on, 106

LK7, 410, 411t
 Load locks, 439
 Logic FEOL integration schemes, 240–241
 Logic technology, 239–241
 Los Alamos National Laboratory (LANL), 476–477
 Low electronic conductivity, 585
 Low ionic conductivity, 585
 Low mechanical force, 254–255
 Low solubility, 585
 Low-energy electron diffraction method (LEED method), 528–529, 531–532
 spot profile analysis *s.*, 528–529
 Low-*k* dielectric(s), 8
 constant, 4
 films, 283
 materials, 126
 Low-*k* materials, 195
 Low-pressure chemical vapor deposition (LPCVD), 583–584
 Low-temperature photochemical reactions, 55
 LPCVD. *See* Low-pressure chemical vapor deposition (LPCVD)
 LPDs. *See* Light point defects (LPDs)

M

Macromolecular organic compound, 597
 Magnetic sector instruments, 722–723
 Magnetically enhanced plasma reactor (MERIE), 403
 Magnetron sputter deposition (MS), 583
 Manufacturing concerns, 243–244
 cost, 243
 ESH, 244
 Marangoni drying. *See* Surface tension drying
 Marangoni effect, 224
 Marangoni IPA dryers or IPA-assisted spin-based Rotagoni dryers, 286
 Mass flow controllers (MFC), 423, 693
 Mass transport processes, 96
 Materials Research Society (MRS), 7
 Maugis and Pollock models (MP models), 174–175
 Mayadas–Shatzkes model, 580
 MBE deposition. *See* Molecular beam epitaxy deposition (MBE deposition)
 MBTA. *See* Methyl-benzotriazole (MBTA)
 Mean free path (λ), 162
 Medium pressure plasma jets, 419
 Mega-ohms (M Ω), 646

Megasonic
 capability, 230
 cleaning, 206–207, 269–270, 285–286
 modules, 289–290
 damage, 126–127
 mechanism, 181
 energy, 126–127, 206
 loss, 290
 particle removal, 206–209
 processing, 459–460
 rinsing, 27
 systems, 26
 tanks, 181
 wet cleaning, 600
 Memory technology, 241
 MEOL. *See* Mid end of line (MEOL)
 Mercury (Hg), 336–337
 MERIE. *See* Magnetically enhanced plasma reactor (MERIE)
 Metal oxide semiconductor (MOS), 388–389, 507–508
 Metal-organic chemical vapor deposition (MOCVD), 583–584
 Metallic contamination, 52–53, 89, 106–120, 565. *See also* Atmospheric molecular contamination (AMC); Particle contamination
 from aqueous media, 106–107
 behavior, 112–116
 metals in dielectric materials, 115–116
 metals in silicon, 113–115, 113f–114f
 effects, 116–120
 bulk silicon defects, 117–118, 118f
 defects in dielectrics, 119–120
 on process control, 116
 metallic ion exchange reactions, 107–110
 metallic redox reactions, 110–112
 removal, 211–213. *See also* Organic contamination removal; Particle removal
 Metallic/metals, 397, 403, 405
 CMP slurries, 270
 complex, 107
 composites, 398–399
 contaminants, 12–13
 in dielectric materials, 115–116
 etch residue removal, 34
 gate technology, 384
 hydroxides, 212
 impurities, 113
 ion exchange reactions, 107–110
 material, 89

- metallic-based resists, 198
 - post-CMP cleaning, 266–267
 - redox reactions, 110–112
 - in silicon, 113–115, 113f–114f
 - surface chemical composition and morphology
 - aluminum, 581–590
 - cobalt, 606–609
 - copper, 590–600
 - properties of interconnect materials, 580t
 - tungsten, 600–606
 - tweezers, 28
 - Metallization, 581–582
 - Methanol (CH_3OH), 15, 30, 357
 - Methyl groups (CH_3), 488
 - Methyl-benzotriazole (MBTA), 596–597
 - N*-Methyl pyrrolidone (NMP), 193
 - Methylsilsequioxane (MSQ), 492–493
 - MFC. *See* Mass flow controllers (MFC)
 - Micelles, 203–204
 - MICRO*, 7
 - Microcontamination. *See MICRO*
 - Microelectronic
 - cryogenic cleaning application examples in microelectronic manufacturing, 472–474
 - fabrication, 579
 - Microetching, 14–15
 - nonuniform, 52–53
 - solutions, 20
 - Microfacets, 544–545
 - Micrometer-and submicrometer-size H_2O droplets, 223
 - Microprocessors, 95–96
 - Microroughening of Si surface, 52–53
 - Microscopic mechanism, 516
 - MicroSiemens (μS), 645–646
 - Microstreaming, 207
 - Microwave (MW), 382–383, 387, 387t
 - spectroscopy, 626–627
 - Mid end of line (MEOL), 8
 - Middle of line (MOL), 8–9, 187, 384
 - post-CMP cleaning, 264–272
 - MIR IR spectroscopy. *See* Multiple internal reflection IR spectroscopy (MIR IR spectroscopy)
 - Miscellaneous cleaning solutions, 20
 - Mixed fluid jet, 209–210
 - mixed fluid jet cleaning, particle sizes distribution with, 209–210, 210f
 - Mixed fluid nozzle-based cleaning, 209–210, 210f
 - ML. *See* Monolayer (ML)
 - Mobile ions, 115–116
 - MOCVD. *See* Metal-organic chemical vapor deposition (MOCVD)
 - Modern Si technology, 508
 - Moisture, 89, 125–126, 126f
 - MOL. *See* Middle of line (MOL)
 - Molecular beam epitaxy deposition (MBE deposition), 344
 - Molecular compounds, 10
 - Molecular contaminant films, 12
 - Molecular contamination measurement, 694
 - Molecular light scattering, 679
 - Molecular vibrations, 628–629, 630f
 - Momentum transfer, 421
 - Momentum transport processes, 96
 - Monoalkyl dimethylchlorosilanes, 488–490
 - Monohydrides, 531, 539–540
 - Monolayer (ML), 338–339, 520
 - Monte Carlo method, 582–583
 - Moore's law, 353
 - MOS. *See* Metal oxide semiconductor (MOS)
 - MOS field effect transistor (MOSFET), 30, 398–399
 - MP models. *See* Maugis and Pollock models (MP models)
 - MRS. *See* Materials Research Society (MRS)
 - MS. *See* Magnetron sputter deposition (MS)
 - MSQ. *See* Methylsilsequioxane (MSQ)
 - Multi-step sputtering, 135
 - Multibubble sonoluminescence, 208
 - Multiple internal reflection IR spectroscopy (MIR IR spectroscopy), 529
 - Multiple wafers, 428–429
 - Multiplexing arrangement, 640–641
 - Multivariate analysis, 628
 - MW. *See* Microwave (MW)
 - MYD model, 176
 - $\text{M}\Omega$. *See* Mega-ohms ($\text{M}\Omega$)
- ## N
- NAND, 241
 - Nano-size slurry particles, 259–260, 259f
 - correlation between electrical short at metal 2 layer and, 260f
 - SEM images of defects, 260f
 - Nanoporous organo-silicate low-k dielectrics, 408
 - Nanostructure formation, 359–361

National Institute of Standards and Technology standards (NIST standards), 650

Native oxide, 328–331, 416–417

Near-atmospheric pressure plasmas, 438 torches, 438

Near-infrared (NIR)

- absorption spectroscopy, 654
- instrumentation for semiconductor fabs, 640
- factors that interfere with measurements, 640–641
- recalibration, 641–645
- response time, 641

spectroscopy, 622

- of complex chemical mixture, 631–632, 634f

Near-regular solution approach, 323–324

Negative binomial yield model, 94

Negative photoresist layer, 316–317

Negatively charged oxygen species, 382–383

Nernst equation, 130, 647, 649

Neutral beams, 448

Neutral solutions for tungsten post-CMP cleaning, 278–279

$(\text{NH}_4)_2\text{SiF}_6$ salts, 335–336

NH_4F etching reactions, 548

NIST standards. *See* National Institute of Standards and Technology standards (NIST standards)

Nitrogen (N_2), 467–468

- carrier gas flow rate and liquid flow rate, 292
- drying, 27
- gas for drying, 356
- nitrogen-based cryogenic aerosols, 458

NMOS, 240–241

NMP. *See* *N*-methyl pyrrolidone (NMP)

Nondispersive absorption, 636–638

Nondispersive instrument, 635, 643f

Non-SC-1 alkaline solutions for FEOL post-CMP cleaning, 276–278

Nonuniform areas of oxide, 356

Nonvolatile residue monito r (NRM), 690–691, 691f

Nonvolumetric particle counters, 678–679

Nozzle-based cleaning, 209–210

NRM. *See* Nonvolatile residue monito r (NRM)

Nucleation mode particles, 101

Nucleophilic attack, 554

O

O_3 and UV/O_3 for organic removal, resist stripping, and surface oxidation, 336–347

- Auger spectra of evaporated Al film on Si surface, 339f
- distance between sample and UV source, 342–343
- exposure types vs. cleaning times, 340t
- growth of SiO_2 on Si surface, 338f
- organic contamination removal, 343–344
- principal wavelengths of low-pressure Hg discharge lamps, 341t–342t
- surface oxidation with UV/O_3 , 346–347
- surface/interface effects, 347
- UV/ozone cleaning, 337–340
 - mechanism, 344–345, 344f
 - in vacuum systems, 345–346
- wavelengths emitted by UV sources, 340–342

OCP. *See* Open circuit potential (OCP)

OES. *See* Optical emission spectroscopy (OES)

OH^- . *See* Hydroxide ions (OH^-)

Ohmi Clean, 21–22, 22f

Ohm's law, 646

OHP. *See* Outer Helmholtz plane (OHP)

Onium salts, 423

Opaque materials, 444

Open circuit potential (OCP), 609

Optical absorption

- instrumentation, 635–636
 - dispersive absorption, 638
 - Fourier transform, 639–641
 - nondispersive absorption, 636–638
- spectroscopy, 620, 628
 - Beer–Lambert law, 629–630
 - calibration, 631–633
 - complex mixtures, 631–635
 - molecular vibrations and wavelength ranges, 628–629
 - near-infrared instrumentation for semiconductor fabs, 640
 - photoresist stripper, 641–644

Optical emission spectroscopy (OES), 441–444

Optical interferometry, 444

Optical particle counters (OPCs), 662, 677–690, 678f

- applications of particle monitoring in liquids, 687–688

- calibration and particle counter
 - verification, 687
 - chemical distribution systems and recirculating baths, 689–690
 - considerations for, 680–684, 683f
 - particle counter performance, 684–687
 - ultrapure H₂O distribution systems, 688–689
 - Optical spectroscopy, 622
 - Optimal brush compression, 285
 - Organic
 - BARC films, 403
 - compounds, 702–703
 - contaminants, 493–494
 - material, 89
 - solvents, 15
 - Organic contamination, 89
 - analysis, 730
 - removal, 197–201, 343–344. *See also*
 - Metallic contamination—removal;
 - Particle removal
 - Organic surface contamination. *See also*
 - Inorganic contamination analysis techniques
 - FTIR, 733–735
 - organic contamination analysis, 730
 - thermal desorption gas chromatography mass spectrometry, 735–736
 - TOF–SIMS, 731–733
 - Organochemical vapor-phase cleaning
 - processes, 31–32, 60
 - Organosilicate glasses (p-OSG), 488–490
 - Original RCA cleaning process, 17–18
 - Outer Helmholtz plane (OHP), 100
 - Overflow rinsing, 232
 - Oxalic acid, 605–606
 - oxalic acid–based cleaning solutions, 279–280
 - Oxidation, 5, 604
 - of hydrogen-terminated silicon surfaces, 549–550
 - oxidation/reduction potentials of metals, 132t
 - in plasma, 61–62
 - plasma cleaning, 416
 - Oxide etching
 - dry methods, 335–336
 - and removal, 191–192
 - Oxide thinning, 119
 - Oxidized Si, 99–100
 - Oxidized surfaces, 506–509
 - chemical composition, 509–513
 - contamination issues with, 517–518
 - structures and morphology, 513–517
 - Oxidizer, 604
 - Oxidizing
 - agents, 621
 - chemistries, 423–424
 - oxidizing-based chemistry, 196–197
 - Oxy-1,4-phenylene-oxy-1,4-phenylene-carbonyl-1,4-phenylene. *See* Polyetheretherketone (PEEK)
 - Oxygen (O₂), 61–62, 424, 584–585
 - exposures, 554
 - interaction with hydrogen-terminated silicon surfaces, 552–557
 - ionization, 552
 - MW plasma emissions, 442
 - O₂-based plasma, 401, 405–406
 - removal, 386
 - O₂-based process, 421
 - Ozone (O₃), 190–191, 476, 509, 511–512, 598
 - cleaning for removal of organics, 30–31
 - ozonated deionized water photoresist stripping, 200–201
 - ozonated solutions, 19–20
 - Ozonization, 20
- P**
- p-channel MOS (PMOS), 240–241
 - p-OSG. *See* Organosilicate glasses (p-OSG)
 - PAC. *See* Photoactive compound (PAC)
 - Parallel plate
 - plasma sources, 437–438
 - reactors, 430
 - Parent molecule, 731
 - Partial least squares (PLS), 629
 - Particle adhesion, 154
 - Brownian motion, 159–160
 - capillary condensation, 171–173, 172f–174f
 - effect of deformation on adhesion, 174–176
 - DLVO theory, 167
 - double-layer electrostatic force, 163–166, 163f
 - drag and lift forces, 154–157
 - electrophoresis, 158–159
 - electrostatic forces, 168–170
 - force measurements, 176–179
 - adhesion force measurement by AFM, 177–178

Particle adhesion (*Continued*)

- measured adhesion forces in liquid media, 175f, 178–179, 179f
- gravitational force, 158
- particle deposition mechanisms in gas and liquid media, 154
- photophoresis, 166–167
- short-range forces, 171
- thermophoresis, 160–162
- turbulence effect, 167
- van der Waals Force, 168
- Particle contamination, 136. *See also*
 - Atmospheric molecular contamination (AMC); Metallic contamination
- effects of particulate contamination, 104–106
- measurements, 695–696
- origins, 101–104
- particle detection
 - in liquids, 696
 - on wafers, 695
- particle volume density, 103f
- Particle monitoring
 - of chemical distribution systems, 689
 - in liquids applications, 687–688
 - on wafer for cleaning tools, 664–668
 - of plasma equipment and vacuum equipment, 667–668
 - of wet stations, 665–667
- Particle removal, 180–183, 201–211, 487–488. *See also* Metallic contamination—removal; Organic contamination—removal
- damage-free megasonic particle removal, 182–183
- effects of cleaning chemistry on Si surface
 - contact angle, 204t
- efficiency and comparison to damage to sized features from cleaning, 212f
- fluid jet, 209–210
- for HF-last Si surfaces using mixed fluid jet spray, 211f
- mechanism
 - of brush scrubbing, 274, 274f
 - in cryogenic aerosol cleaning, 467–471
 - by HF chemical only, 274–275, 275f
- megasonic, 206–209
 - damage mechanism, 181
- monitoring, 210–211
- particle addition to wafer after steps in wet cleaning process, 203f

- RCA cleaning and HF, 202–204
- SC-1, 201–202
- and surface roughness, 204–206
- Particle removal efficiency (PRE), 204–205, 210–211, 467, 474f, 663
- Particle size distribution (PSD), 675
- Particle(s), 11, 13
 - counter
 - performance, 684–687, 685f–686f
 - verification, 687, 688f
 - and defects measurements
 - cleaning process steps and equipment requiring particle monitoring, 661–662
 - particulate contamination and defect analysis, 660–661
 - particulate contamination and defect measurement equipment, 661
 - deposition, 154
 - Brownian motion, 159–160
 - DLVO theory, 167
 - double-layer electrostatic force, 163–166, 163f
 - drag and lift forces, 154–157
 - electrophoresis, 158–159
 - gravitational force, 158
 - mechanisms in gas and liquid media, 154
 - photophoresis, 166–167
 - thermophoresis, 160–162
 - turbulence effect, 167
 - detection, 684–685
 - detection on wafers, 695
 - measurement
 - in air and gas, 691–694
 - in liquid chemicals, 677–691
 - scan maps of dielectric films, 283, 283f
 - size, 675, 683f
 - transport, 154–167
- Particulate contaminants
 - defect and particle measurements on wafers, 662–677
 - measurements of particles and defects, 660–662
 - molecular contamination measurement, 694
 - particle contamination measurements, 695–696
 - particle measurement
 - in air and gas, 691–694
 - in liquid chemicals, 677–691
 - Particulate contamination, 89, 720
 - back surface particles, 104–106

- effects, 104–106
- front surface particles, 104
- particles on lithographic media, 106
- Partitioning testing, 665–666
- Passivation
 - challenges, 262–264
 - layer, 581
 - process and contamination issues, 581, 607–609, 608f
 - tungsten process and contamination issues, 604–606, 605f
- Pattern collapse, 128, 227–228
- Patterned device wafers, 668
- Patterned photoresist, 33
- Patterned wafer defect inspection, 672–673
- PCA. *See* Principal component analysis (PCA)
- PEEK. *See* Polyetheretherketone (PEEK)
- PER. *See* Postetch residue (PER)
- Per-wafer-pass measurements (PWP), 665
- Perfluoro compounds (PFC), 136–137, 385
- Perfluoroalkoxy components (PFA components), 134, 228–229
- Peroxy-titanate ($\text{TiO}_2 \cdot \text{H}_2\text{O}_2$), 197
- Peroxydisulfuric acid ($\text{H}_2\text{S}_2\text{O}_8$), 16–17
- PFA. *See* Polyfluoroalkoxy resins (PFA)
- PFA components. *See* Perfluoroalkoxy components (PFA components)
- PFC. *See* Perfluoro compounds (PFC)
- pH, 620, 622
 - effect, 169, 170f
 - electrode, 648–649
 - solution, 593
- Phosphorus (P), 319–320
- Photo-induced
 - corrosion, 271–272
 - of Co at p and n electrodes, 273f
 - micrographs, 273f
 - mechanisms, 552
- Photoacid generators, 198, 423
- Photoactive compound (PAC), 409–410, 423
- Photochemically enhanced cleaning process, 337
- Photoelectric effect, 715
- Photoelectron, 715
- Photophoresis, 166–167
 - radiation pressure, 166–167
 - radiometry, 166
- Photophoretic force, 166
- Photoresist(s), 33, 122, 134, 190–191, 398–399, 480–485
 - compounds, 198
 - removal, 190–191
 - chemistries for, 422–426
 - stripper, 641–644
 - DOE, 644–645
 - in-line monitoring, 645–653
 - and residue removers, 625
 - stripping, 197–201
 - chemical oxides, 200f
 - ozonated deionized water, 200–201
 - process, 422
 - sulfuric acid/hydrogen peroxide, 198–200
 - temperature change of concentration SPM solutions, 199f
- Photosensitized oxidation processes, 344
- Physical contact, 138
- Physical contaminants, 313
- Physical vapor deposition (PVD), 241–242, 582
- Physisorption, 609
- Pipe defects, 336
- Piranha
 - etch, 16
 - solution, 275–276
- Planarity, 259
- Planarization, 604
- Planck's equation, 626
- Plasma stripping, cleaning, and surface
 - conditioning, 32–33, 60, 380–381, 385–386, 430
 - applications, 385–419
 - bulk photoresist stripping, 386–389
 - cleaning and conditioning plasma chambers, 419
 - other applications of and approaches to plasma cleaning, 418–419
 - postetch aluminum stripping and cleaning, 412–413
 - postetch resist stripping and cleaning, 397–412
 - postion implant resist stripping and cleaning, 390–397
 - pre-high-k deposition and oxidation plasma cleaning, 416
 - preepitaxial deposition cleaning, 413–416
 - premetal and predielectric deposition plasma cleaning, 416–418
 - challenges, 383–385
 - chemistries for, 421–427, 421t
 - cleaning and surface conditioning, 426

- Plasma stripping, cleaning, and surface conditioning (*Continued*)
 for photoresist and residue removal, 422–426
 pressure, 427
 temperature, 426–427
 cryogenic aerosol
 cleaning and conditioning, 35
 and supercritical fluid cleaning, 34–35
 difference between stripping and cleaning, 382f
 equipment, 428–440
 auxiliary equipment and components, 438–439
 commercially available plasma equipment, 430–438
 downstream microwave plasma sources, 431–433
 downstream radio frequency plasma sources, 431
 ECR plasma sources, 438
 inductively coupled plasma sources, 433–436
 other plasma sources, 438
 parallel plate plasma sources, 437–438
 remote plasmas, 429–430
 safety, 440
 mechanisms
 physical and chemical mechanisms for, 420–421
 plasma parameters, 419–420
 plasma damage, 445–448
 plasma diagnostics, 440–445
 plasma processing, 380–383
 predeposition cleaning, 34
 removal of etch residues, 33–34
 stripping of bulk photoresist, 33
 supercritical fluid cleaning, 36
 surface conditioning, 34
 Plasma(s), 381–383
 ashing, 32–33
 cleaning and conditioning plasma chambers, 419
 damage, 445–448
 deposition, 381
 diagnostics, 440–445
 and endpoint detection for stripping and cleaning, 440–441
 laser and optical interferometry, 444
 OES, 442–444
 residue gas mass spectroscopy, 444–445
 equipment, 667–668
 etching, 33–34, 397
 jet cleaning, 418
 parameters, 419–420, 420t
 plasma-assisted stripping applications, 33
 plasma-enhanced CVD, 583–584
 process, 190–191, 380–383, 428
 reactors, 136–138
 particle contamination, 136
 post-etch residue and resist, 136–138
 types, 33
 PLS. *See* Partial least squares (PLS)
 PMOS. *See* p-channel MOS (PMOS)
 Point of zero charge (PZC), 99–100
 Point-of-use (POU), 14, 133
 filtration, 236–237
 Poisson distribution, 93
 Poisson equation, 101, 163–164
 Polar molecule, 317
 Polar silanol group, 492
 Polarized IR absorption spectroscopy, 542–543
 Polarized spectra, 529
 Polish byproducts, 261, 261f
 Polisher platen, 294–295
 Polishing process, 254
 Polycrystalline material, 581–582
 Polycrystalline Si (polySi), 8
 Polycrystalline β -W films, 601–602
 Polyetheretherketone (PEEK), 228–229
 Polyfluoroalkoxy resins (PFA), 236
 Polymer, 403–404
 formation, 397
 removal, 397–412, 480–485
 cleaning postetch contact, via, and trench polymer, 403–412, 411t
 cleaning postetch polysilicon gate polymer, 398–403
 Polypropylene (PP), 134
 polySi. *See* Polycrystalline Si (polySi); Polysilicon (polySi)
 Polysilicon (polySi), 191, 257–258. *See also* Silicon(Si)
 emitters, 361
 films, 460
 gate cleaning, 434
 gate deposition, 362–363
 Polystyrene latex (PSL), 674–675, 676f
 Polytetrafluoroethylene (PTFE), 121–122, 134, 228–229
 Polyvinyl acetal disks (PVA disks), 256, 286

- brush scrubbing module, 290–291, 291f
- cleaning module, 290–291
- Polyvinyl alcohol (PVA), 606
 - brush scrubbing, 606
- Popping
 - process, 391–392
 - residue, 394–395
- Pore-sealing, 409, 495–496
- Porous Si–O–C films, 488–490
- Positively charged oxygen species, 382–383
- Post-CMP cleaning. *See* Postchemical
 - mechanical planarization cleaning (Post-CMP cleaning)
- Post-plasma stripping residue removal, 485–486
- Postchemical mechanical planarization
 - cleaning (Post-CMP cleaning), 254–264, 294–295
 - application of cleaning principle for, 257–259
 - challenges, 255–256
 - chemicals, 273–285
 - acidic and non-SC-1 alkaline solutions for FEOL post-CMP cleaning, 276–278
 - aluminum metal gate post-CMP cleaning solutions, 279–280
 - ammonium hydroxide and neutral solutions for tungsten post-CMP cleaning, 278–279
 - copper/low-k dielectric post-CMP cleaning solutions, 280–285
 - diluted NH_4OH , TMAH, SC-1, and HF for FEOL post-CMP cleaning, 274–276
 - potential–pH diagram of tungsten–water system, 278f
- defectivity and passivation challenges, 262–264
- dielectric, 264–266
- drying challenges, 261–262
- equipment, 285–293
 - brush scrubbing cleaning modules, 287–289
 - drying modules, 292–293
 - jet spray cleaning module, 291–292, 291f
 - megasonic cleaning modules, 289–290
 - PVA disk cleaning module, 290–291
- FEOL, MOL, and BEOL, 264–272
- future direction, 293–296
- interconnect, 267–272
- metal, 266–267
 - of new materials, 295–296
 - requirements for, 257t
 - sequence, 257f
 - slurry particles removal and polish byproducts, 261
 - STM image, 513–514
 - zeta potential, 257–258, 258f
- Postetch
 - aluminum stripping and cleaning, 412–413
 - cleaning, 191
- Postetch residue (PER), 192, 195
 - polymers removal stripping and cleaning, 397–412
 - and resist, 136–138
- Postimplant species residue, 394–395
- Postion implant resist stripping and cleaning, 390–397, 392t–393t, 394f
- Postmetal etching resist or residue, 20
- Postpolishing on-platen chemical buff, 279
- Postprocess cleaning, 188–191, 190t
- Poststrip cleaning, 385–386
- Potassium permanganate (KMnO_4), 621
- Potassium sorbate, 596–597
- Potentiodynamic polarization, 595
- Potentiometric conductivity measurements, 647
- POU. *See* Point-of-use (POU)
- Pourbaix diagram(s), 106–107, 107f, 258–259
 - of Al– H_2O system, 279–280, 279f, 586, 586f
- Power law size distribution, 102–103
- PP. *See* Polypropylene (PP)
- PRE. *See* Particle removal efficiency (PRE)
- Pre-epitaxial deposition cleaning, 356–357, 413–416
- Pre-high-k deposition, 416
- Pre-thin film measurement, 361–362
- Pre-tungsten CVD cleaning, 357–358
- Precipitation of metals, 114
- Precleaning. *See* Preprocessing cleaning
- Precursors, 607
- Predeposition
 - cleaning, 34, 381
 - surface cleaning, 380
- Predielectric deposition plasma cleaning, 416–418
- Preferential etching mechanism, 545–549
 - Si (100) etching mechanism, 548–549
 - Si (111) surfaces mechanism, 546–548
- Prefurnace cleaning, 354–356

Pregate dielectric cleaning and surface conditioning, 34

Premetal deposition plasma cleaning, 416–418

Preprocessing cleaning, 187–188

Pressure, 427, 583, 601

- pressure–enthalpy diagram of CO_2 , 465, 465f
- pressure–temperature diagrams, 478–479

Preston equation, 254

Principal component analysis (PCA), 629, 631

Principle of mass conservation, 96

Process control applications, 702

Process equipment, 362–369

- anhydrous HF etching equipment, 363–364
- HF/ NH_3 processing equipment, 368–369
- integrated HF vapor system, 364f
- photochemical etching of Si, 367f
- subatmospheric gas-phase HF etching system, 364–366, 365f
- UV/ O_3 processing equipment, 367–368

Process gases, 133–134

Process liquids, 134

Process variables affecting cleaning, 214–221

- concentration and temperature effects, 215–216
- decomposition rate of H_2O_2 , 218f
- decrease of NH_4OH concentration, 217f
- deionized water quality, 219–221
- sequence effects, 219
- time and aging effects, 216–219

Process-induced microscratches, 606

PSD. *See* Particle size distribution (PSD)

PSL. *See* Polystyrene latex (PSL)

PTFE. *See* Polytetrafluoroethylene (PTFE)

Purge gas, 467–468

PVA. *See* Polyvinyl alcohol (PVA)

PVA disks. *See* Polyvinyl acetal disks (PVA disks)

PVD. *See* Physical vapor deposition (PVD)

PWP. *See* Per-wafer-pass measurements (PWP)

Pyrex filter, 337–338

PZC. *See* Point of zero charge (PZC)

Q

QDR. *See* Quick dump rinser (QDR)

Quality control (QC), 713–714

Quartz (SiO_2). *See* Silicon dioxide (SiO_2)

Quick dump rinse, 27

Quick dump rinser (QDR), 231–232, 232f

R

Radiation

- damage, 387
- pressure, 166–167

Radio frequency (RF), 310, 382–383, 723–724

- plasma emissions, 442
- RF-biased plasmas, 437
- sources, 430–431

Radioactive

- ^{18}F , 520
- isotopes, 38
- Na-22, 38–39

Radiochemical studies of surface

- contamination, 38–46, 38f
- development of original RCA wafer cleaning procedure, 45–46

Radiometry, 166

Radionuclides. *See* Radioactive isotopes

Rapid optical surface treatment, 361–362

Rapid thermal annealing (RTA), 607

Rare gases, 463

Rate of evaporation, 320–321

Rayleigh scattering theory, 674, 682

Rayleigh streaming, 207

RCA

- 2N1482pnp, 38–39
- 2N217nnp, 38–39
- cleaning, 202–204, 569
- modifications of RCA cleaning process, 19
- sequence, 187–188
- development of original RCA wafer cleaning procedure, 45–46
- RCA-type wet chemical treatment, 414
- Standard Clean, 17, 58, 506

Reactive fragments, 314

Reactive ion etching (RIE), 188, 337, 437

Reactive species, 429–430

Reactor, 428

Real-time particle monitoring of liquid chemicals, 677

Recalibration, 641–645

Recirculating baths, 689–690, 690f

Rectified diffusion, 206–207

Reducing chemistries, 424

Reference methods, 634

Refractometry, 620–621

Reinberg reactor, 428

- Relative humidity, 125
 - Relative sensitivity factor (RSF), 721–722
 - Relative standard deviation (RSD), 685
 - Reliability
 - classification according to, 92
 - relationship to defectivity, 92–96
 - defect size distribution, 95
 - effective area, 95–96
 - statistical yield models, 93–94
 - Reliable detection of particles, 696
 - Remote ECR plasma cleaning
 - process, 417
 - Remote H₂-based plasmas, 416
 - Remote plasma(s), 429–430
 - cleaning of surfaces, 420
 - photoresist stripping, 430–431
 - Remote reactors, 429
 - Reoxidation, 329
 - Repeatability test (RPT test), 675–677
 - Replacement metal gate (RMG), 254
 - Reproducibility test (RPD test), 676–677
 - Residual Cl species, 192
 - Residue
 - formation, 331–335
 - gas mass spectroscopy, 444–445
 - removal, 380, 422–426, 381
 - Resist
 - materials, 386
 - removal, 387
 - stripping, 188–191
 - Resistance (*R*), 646
 - Resistivity, 620–622, 645–651
 - of material, 580
 - measurements, 648–649
 - calibration of electrode measurements, 650
 - installation of electrodes, 650–652
 - effect of temperature on electrode measurements, 649–650
 - ultrapure water resistivity measurements, 652–653
 - Response time, 641
 - Reverse micelle, 485
 - RF. *See* Radio frequency (RF)
 - RF reactive ion etching (RIE), 395
 - RI. *See* Index of refraction (RI)
 - RIE. *See* Reactive ion etching (RIE); RF reactive ion etching (RIE)
 - Rinsing, 186–187, 221–228
 - defects relating to, 225–228
 - liquid between two pattern lines, 228f
 - pattern collapse, 227–228
 - watermarks, 225–227
 - evaluation of rinsing performance, 221–223
 - process, 651
 - processors, 231–234
 - dump rinsing, 231–232
 - effects of rinse tank configuration, 233f
 - overflow rinsing, 232
 - spin-spray rinsing, 233–234
 - RMG. *See* Replacement metal gate (RMG)
 - RMS. *See* Root mean square (RMS)
 - Rocking scan, 720
 - Root mean square (RMS), 130, 513–514, 737
 - Rotagani, 28
 - drying module, 293, 293f
 - system, 59
 - RPD test. *See* Reproducibility test (RPD test)
 - RPT test. *See* Repeatability test (RPT test)
 - RSD. *See* Relative standard deviation (RSD)
 - RSF. *See* Relative sensitivity factor (RSF)
 - RTA. *See* Rapid thermal annealing (RTA)
 - Ruthenium (Ru), 4, 241–242, 270, 495–496
- ## S
- SA-80, 16–17
 - SAC. *See* Semiaqueous chemistry (SAC)
 - Sacrificial oxide strip, 362–363
 - Sapphire (Al₂O₃), 430–432
 - ceramic, 430
 - optical materials, 627
 - SC-1. *See* Standard clean 1 (SC-1)
 - Scanning collection solution (SCS), 706
 - Scanning electron microscopy (SEM), 138, 259, 399–400, 403, 473–474, 485, 486f, 588f, 595, 603f, 661, 728
 - Scanning spectrometer, 636
 - Scanning surface inspection systems (SSIS), 675
 - Scanning transmission electron microscopy, 592, 592f
 - Scanning tunneling microscope (STM), 513–514, 531–532
 - Scattering-based defect inspection, 670
 - scCO₂ for resist removal process (SCORR process), 476–477, 480–481
 - scCO₂. *See* Supercritical carbon dioxide (scCO₂)
 - SCF. *See* Supercritical fluid (SCF)
 - SCFM. *See* Standard cubic feet per minute (SCFM)
 - Schlichting streaming, 207
 - SCO₂. *See* CO₂-snow (SCO₂)

- SCORR process. *See* scCO_2 for resist removal process (SCORR process)
- Scratch-related alumina residue, 279–280, 280f
- Scratches, 93
- SCS. *See* Scanning collection solution (SCS)
- SD. *See* Standard deviation (SD)
- Secondary ion mass spectroscopy (SIMS), 27–28, 329, 414–415, 720–724
- Segregation gettering, 115
- Self-annealing, 592
- SEM. *See* Scanning electron microscopy (SEM)
- Sematech Surface Preparation and Cleaning Conference*, 7
- SEMI Standard F21–95, 120
- Semiaqueous chemistry (SAC), 590
- Semiconductor
- contamination properties in semiconductor environment, 702–703
 - device production, 702
 - equipment and materials international, 7
 - fabrication process, 581
 - industry, 620
 - wafer types, 11
- Semiconductor Pure Water and Chemicals Conference*, 7
- Sequence effects, 219
- Shallow trench isolation (STI), 241, 254, 353
- Shockley-Read-Hall
- recombination lifetime, 117
 - theory, 117
- Short-range forces, 171
- Short-wave ultraviolet, 30–31
- Si on insulator (SOI), 308–309
- SiC. *See* Silicon carbide (SiC)
- SiConi process, 336
- SiGe. *See* Silicon germanium (SiGe)
- Sight sputtering, direct line of, 135
- Signal averaging, 640
- Silanol (SiOH), 318, 331–333
- SiLi detector. *See* Lithium-drifted silicon detector (SiLi detector)
- Silica (SiO_x), 474
- etching, 494–495
 - particle, 474
 - silica-alumina ceramic, 430
- Silicon (Si), 4, 187, 307, 581–582
- back-bond polarization, 554
 - carbides, 564–565
 - etching rate, 216, 216f
 - metals in, 113–115, 113f–114f
 - molecular beam epitaxy, 514
- Si (100)
- etched in aqueous HF solutions, 528–531
 - etched in buffered HF solutions, 537–545
 - etching mechanism, 548–549
- Si (111)
- etched in aqueous HF solutions, 528–531
 - etched in buffered HF solutions, 531–537
 - surfaces mechanism, 546–548
- Si-rich salts, 336
- surfaces analytical measurement, 702
- Silicon carbide (SiC), 12
- Silicon dioxide (SiO_2), 8–9, 188, 257–258, 430–432, 712
- crystal microbalance, 586–587
 - etching films, 15
 - optical materials, 627
 - quartz-halogen lamps, 439
 - residue, 394–395
- Silicon germanium (SiGe), 4, 296
- alloys, 307
 - post-CMP cleaning, 296
- Silicon nitride (Si_3N_4), 8, 264, 472–474
- particle removal from silicon dioxide film, 288–289, 289f
- Silicon oxidation, 405–406
- Silicon oxide(s), 328–331
- CMP, 264
 - removal with HF vapor, 316–336, 317f
 - bulk etching, 319–331
 - dry methods of oxide etching, 335–336
 - etching rate, 325–328
 - initiation, 318–319
 - moisture content, 334f
 - partial pressures of HF and H_2O , 325f
 - response surface plot of thermal SiO_2 etching rate, 320f
 - termination, residue formation, surface roughness, and electrical performance, 331–335
 - transport and reaction processes in SiO_2 etching, 322f
 - vapor pressures of HF and H_2O with changes, 324f
- silicon oxide-terminated surfaces
- chemical composition of oxidized surfaces, 509–513

- contamination issues with oxidized surfaces, 517–518
- oxidized surfaces, 506–509
- structures and morphology of oxidized surfaces, 513–517
- Silicon tetrachloride (SiCl_4), 315–316
- Silicon wafer cleaning technology, 5
 - evolution of wafer cleaning science and technology, 36–60
 - importance of clean and conditioned wafer surfaces, 5–9
 - modern cleaning and surface conditioning science and technology
 - future of cleaning and surface conditioning, 63
 - trends and milestones 2007 to 2017, 61–63
 - wafer cleaning and surface conditioning technology, 14–36
 - wafer contamination aspects, 10–14
- Silicon-on-insulator (SOI), 116, 671–672, 672f
- Simple immersion tank, 26
- SIMS. *See* Secondary ion mass spectroscopy (SIMS)
- Single-wafer
 - brush scrubbers, 26
 - chemical processors, 230–231
 - cleaning, 61–62, 194–195, 234
 - downstream reactors, 430
 - immersion tank megasonic module, 289, 290f
 - process modules, 622–623
 - processing, 24–25, 209–210
 - single-wafer/short-cycle clean, 21, 24–25, 25f
 - SRDs, 286
 - tool for immersion wet processing, 25
 - wet processing module, 638
- SiOH . *See* Silanol (SiOH)
- “Skin-formation”, 121
- Slurry particles, 597–598
 - removal, 261, 261f
- “Snow”, 465
- Sodium (Na), 10, 388–389
- Sodium chloride (NaCl), 645
- Soft etching process, 418
- Soft-silicon etching, 380
- “Soft” etches, 437
- SOI. *See* Si on insulator (SOI); Silicon-on-insulator (SOI)
- Solvent
 - stability, 565–566
 - vapor drying, 27–28
- SOM. *See* Sulfuric acid–ozone mixture (SOM)
- Sonic wave, 206–207
- SPC. *See* Statistical process control (SPC)
- Spectrum of absorption, 629
- Spin drying, 27, 223
- Spin rinse dryers (SRD), 256, 261–262, 292
 - copper/low-k wafer defect performance
 - comparison between IPA drying and, 263f
 - defect scan maps after, 262f
 - Spin-spray rinsing, 233–234
- SPM. *See* Sulfuric acid and hydrogen peroxide mixture (SPM)
- Spray processors, 230
- Sputter etching process, 357, 417
- Sputtered off, contaminants, 135
- Sputtered SiO_2 residue, 394–395
- Sputtering process, 582–583
- SR. *See* Synchrotron radiation (SR)
- SRD. *See* Spin rinse dryers (SRD)
- SSIS. *See* Scanning surface inspection systems (SSIS)
- Stainless steel, 431
- Standard clean 1 (SC-1), 50, 187–188, 459–460, 509, 514–515, 621
 - for FEOL post-CMP cleaning, 274–276
 - particle removal, 201–202
 - process, 517, 569
 - solution, 17–18, 207
- Standard clean 2 (SC-2), 50, 187–188, 212–213, 509, 514–515
 - aqueous solution, 187–188
 - solution, 18
 - treatment of substrate, 50
- Standard cubic feet per minute (SCFM), 693
- Standard deviation (SD), 685
- State-of-the-art
 - fabs, 236–237
 - ICs, 396–397
 - fabrication processes, 380–381
 - patterned wafer inspection systems, 673
 - wafer cleaning processes, 201
- Static closed systems, 26
- Statistical process control (SPC), 664, 665f
- Statistical yield models, 93–94
- Steam test, 338
- “Step flow” etching mechanism, 547–548
- Stern layer, 163, 202

- STG. *See* Surface tension gradient (STG)
- STI. *See* Shallow trench isolation (STI)
- “Stiction”, 359
- Stiction-related pattern collapse, 227
- STM. *See* Scanning tunneling microscope (STM)
- Strained dihydrides, 543–544
- Stripping process, 381–382
- Subatmospheric gas-phase HF etching system, 364–366, 365f
- Submicron colloidal crystal particles of cryogenic gas, 35
- Substrate charging, 129
- Sulfur (S), 518
- Sulfuric acid (H_2SO_4), 120, 188, 275–276 mixtures, 16–17
- Sulfuric acid and hydrogen peroxide mixture (SPM), 8, 137, 188, 391–392, 395, 509, 514–515, 625 photoresist stripping, 198–200
- Sulfuric acid–ozone mixture (SOM), 16–17
- “Sum peak”, 718–719
- Supercritical carbon dioxide (scCO_2), 56, 60, 458, 460–461, 476, 482, 484, 487, 493–494, 496–497 cleaning, 60 critical density, 478–479 microemulsions, 484–485 region, 464 repair of plasma-damaged low-k, 492–493 scCO_2 -based silylation reactions, 497
- Supercritical fluid (SCF), 409, 475–497 cleaning technologies, 34–36, 458–461, 477–480 dielectric repair applications, 488–494 gases properties for, 477–478 particle removal, 487–488 photoresist and polymer removal, 480–485 pore sealing, 495–496 post-plasma stripping residue removal, 485–486 pressure–temperature diagrams, 478–479 supercritical fluid drying and porogen removal, 496 supercritical fluid metal removal, etching, and surface conditioning, 494–495 for surface conditioning and cleaning, 460–461 drying and porogen removal, 496 metal removal, 494–495
- Surface analysis techniques, 528–529
- Surface chemical composition and morphology, 506 contamination and stability of hydrogen-terminated surfaces, 549–565 hydrogen-terminated silicon surfaces, 518–549 silicon oxide–terminated surfaces, 506–518 stability and reactivity of hydrogen-terminated silicon surfaces, 565–568
- Surface cleaning processes, 569
- Surface conditioning, 6, 34, 187, 309, 381 alternative wet-chemical cleaning and surface conditioning systems, 21–25 cryogenic aerosols and supercritical fluids for, 460–461 dry cleaning and surface conditioning processes, 28–36 equipment for implementing wet-chemical cleaning, 26–27 liquid processes and wafer drying techniques, 14–15 supercritical fluid metal removal, etching, and copper removal, 494 etching of silica, 494–495 technology, 14–36 wafer rinsing, drying, and storing, 27–28 wet-chemical cleaning processes, 15–21
- Surface contaminants, 550
- Surface contamination measurement techniques, 154, 705. *See also* Inorganic contamination analysis techniques
- Surface defectivity, 89
- Surface hydrophobic chemicals, 228
- Surface microcontamination, 13
- Surface morphology analysis techniques, 736–737 and composition of aluminum, 581–584 of cobalt, 606–607 of copper, 590–593 of tungsten, 600–603
- Surface oxidation with UV/O_3 , 346–347
- Surface preparation, 56–57, 187
- Surface Preparation Processes for Semiconductors*, 7
- Surface roughness, 204–206, 331–335, 507

Surface science techniques, 531–532
 Surface tension
 drying, 28, 224–225, 292–293, 293f
 effects, 28
 Surface tension gradient (STG), 224–225
 Surface-sensitive technique, 719
 Surface/interface effects, 347
 Surfactant(s), 16
 as additives, 58
 molecules, 203–204
 Synchrotron radiation (SR), 718

T

“T-topping”, 121
 Tafel plots of Cu and Co, 271, 272f
 TAZ. *See* 1,2,4-Triazole (TAZ)
 TCP. *See* Transformer-coupled plasma (TCP)
 TDA–MS. *See* Thermal Desorption Analysis Gas Chromatography Mass Spectrometry (TDA–MS)
 TDDb. *See* Time-dependent dielectric breakdown (TDDb)
 TDMAT. *See* Tetrakis-dimethylamido titanium (TDMAT)
 TE. *See* Thermal evaporation (TE)
 Technical jargon, 8
 Teflon, 403
 TEM. *See* Transmission electron microscopy (TEM)
 Temperature, 583, 586–587, 601, 635, 642
 effects, 215–216
 electrode measurements effect, 649–650
 TEOS. *See* Tetraethylorthosilicate (TEOS)
 Termination, 331–335
 Tetra (hydroxyethyl) ethylenediamine
 chelating agent, 596–597
 Tetraethylorthosilicate (TEOS), 335–336, 495
 Tetrakis-dimethylamido titanium (TDMAT), 495–496
 Tetramethyl ammonium hydroxide (TMAH), 179, 274, 598
 for FEOL post-CMP cleaning, 274–276
 TMAH–H₂O₂ mixtures, 58
 Thermal desorption gas chromatography mass spectrometry, 735–736
 Thermal Desorption Analysis Gas Chromatography Mass Spectrometry (TDA–MS), 707t–709t
 Thermal energy, 314
 Thermal evaporation (TE), 583
 Thermal oxidation and diffusion, 8
 Thermal oxides, 328–331, 507
 Thermal processing, 138–139
 Thermal SiO₂ etching rate, 321
 at anhydrous HF concentrations, 322f
 function of HF concentration, 321f
 response surface plot, 320f
 Thermally grown oxides, 508
 Thermionic vacuum arc method, 601
 Thermophoresis, 160–162
 Time effects, 216–219
 Time-dependent dielectric breakdown (TDDb), 262–263, 263f, 354
 Time-dependent haze, 121
 Time-of-flight secondary ion mass spectrometry (TOF–SIMS), 703, 707t–709t, 714, 724–728, 724f, 731–733
 detection limits, 725t
 separation of isobars at nominal mass, 733f
 Titanium nitride (TiN), 196
 metal hardmask, 196, 198f
 Titanyl (TiO²⁺), 197
 Titration, 620–621, 633
 TMAH. *See* Tetramethyl ammonium hydroxide (TMAH)
 TMCS. *See* Trimethylchlorosilane (TMCS)
 TOC level. *See* Total organic carbon level (TOC level)
 TOF–SIMS. *See* Time-of-flight secondary ion mass spectrometry (TOF–SIMS)
 Total organic carbon level (TOC level), 220, 220f, 666
 Total reflection X-ray fluorescence (TXRF), 329, 704, 707t–709t, 715–719, 715f
 analysis of SiO₂ film, 284–285, 285f
 Trace hydrocarbons, 517–518
 Transformer-coupled plasma (TCP), 403
 Transient effects during rinsing, 127
 Transmission electron microscopy (TEM), 485
 Trialkylchlorosilanes, 488–490
 Trialkylsilanes, 495
 1,2,4-Triazole (TAZ), 609
 Triboelectric charging, 129
 Trimethyl 2-hydroxyethyl ammonium hydroxide, 202
 Trimethylchlorosilane (TMCS), 488–490
 Triple point, 462
 Tunable broadband brightfield illumination technology, 673

- Tungsten (W), 8, 254, 580, 581f, 715–716
 - ammonium hydroxide and neutral solutions for tungsten post-CMP cleaning, 278–279
 - contact CMP, 266, 266f
 - gate CMP, 266, 266f
 - passivation process and contamination issues, 604–606, 605f
 - potential–pH diagram of tungsten–water system, 278f
 - surface morphology and composition of tungsten, 600–603
- Tungsten hexacarbonyl ($W(CO)_6$), 601
- Tungsten oxide (WO_3), 278–279
- Turbulence effect, 167
- Two-step
 - immersion cleaning sequence, 187–188
 - oxidation mechanism, 562
- TXRF. *See* Total reflection X-ray fluorescence (TXRF)

U

- UHV. *See* Ultrahigh vacuum (UHV)
- ULK nanoporous dielectrics. *See* Ultralow k nanoporous dielectrics (ULK nanoporous dielectrics)
- ULPA filters. *See* Ultra-low penetration air filters (ULPA filters)
- ULSI. *See* Ultra-large-scale integration (ULSI)
- Ultra-large-scale integration (ULSI), 5
- Ultra-low penetration air filters (ULPA filters), 89, 103–104
- Ultraclean, 5–6
 - sample vial, 710–711
 - Si, 4
- Ultrahigh vacuum (UHV), 520
- Ultralow k nanoporous dielectrics (ULK nanoporous dielectrics), 406–408
- Ultrapure H_2O (UPW), 20–21, 550–551, 620, 625, 660, 710–712
 - distribution systems, 688–689
 - resistivity measurements, 652–653
- Ultrapure, 14
 - gas delivery, 439
- Ultrasonic tanks, 181
- Ultratrace impurity analysis of wafer surfaces. *See also* Wafer contamination
 - advantages and limitations of surface contamination measurement techniques, 705
 - analytical measurement of silicon surfaces, 702
 - characteristics of analytical techniques, 707t–709t
 - contamination properties in semiconductor environment, 702–703
 - future contamination analysis techniques and trends, 737–738
 - inorganic contamination analysis techniques
 - AES, 728–730
 - EDS, 728
 - SIMS, 720–724
 - TOF–SIMS, 724–728, 724f
 - TXRF spectrometry, 715–719, 715f
 - VPD, 706–714
 - VPD/TXRF, 719–720
 - XPS, 730
 - organic surface contamination
 - FTIR, 733–735
 - organic contamination analysis, 730
 - thermal desorption gas chromatography mass spectrometry, 735–736
 - TOF–SIMS, 731–733
 - selection criteria for ultratrace metrology, 703–705
 - surface morphology analysis techniques, 736–737
- Ultratrace metrology, selection criteria for, 703–705
- Ultraviolet (UV), 408, 622, 671–672
 - cleaning for removal of organics, 30–31
 - exposure, 335
 - irradiation, 219–220, 306, 335
 - lamp, 367–368
 - lamp heating, 286
 - light, 337–338
 - photoemission spectroscopy, 521
 - photons, 445–446
- Ultraviolet/ozone (UV/ O_3), 458
 - cleaning, 306, 337–340, 356
 - mechanism, 344–345, 344f
 - and surface conditioning applications, 59–60
 - in vacuum systems, 345–346
 - exposure vs. oxide thickness on Al, 347t
 - processing equipment, 367–368
 - surface oxidation with, 346–347
- Uncertainty, 675–677
- Uniformity, 259
- Unpatterned wafer defect inspection, 669–672

UPW. *See* Ultrapure H₂O (UPW)
 Uric acid, 596–597
 UV. *See* Ultraviolet (UV)
 UV/chlorine (UV/Cl₂), 31
 dry wafer cleaning, 348f
 mechanisms, 351–352
 for metallic contamination removal,
 336–347, 349
 recovery of cleanliness, 350f
 surface impurity concentrations, 350f
 TXRF data, 348f
 vapor pressure
 for “high”—vapor pressure metals, 351f
 of metal chlorides, 352f
 vapor-phase cleaning for metal removal, 31

V

Vacuum
 equipment, 667–668
 processing, 139–141, 140t
 system, 139, 439
 UV, 408, 412
 van der Waals Force, 168, 174f, 469–470
 van Laar equations, 323–324
 Vapor phase decomposition (VPD), 706–714,
 726
 chamber, 712–713
 contamination sources during, 712–713
 hydrofluoric acid decomposition stage, 710
 instrumental analysis of VPD droplet,
 712–713
 wafer scanning process, 710–711
 collection efficiency of scanning
 process, 711–712
 Vapor Phase Decomposition/Inductively
 Coupled Plasma Mass Spectroscopy (VPD/TXRF),
 706, 707t–709t, 714, 719–720
 Vapor phase decomposition-inductively
 coupled plasma mass spectrometry
 (VPD/ICP–MS), 703, 706, 719
 Vapor-phase
 cleaning processes and methods, 29–32,
 54–56, 59
 HF vapor etching, cleaning, and surface
 conditioning, 29–30
 organochemical vapor-phase cleaning
 processes, 31–32
 ultraviolet/ozone cleaning for removal of
 organics, 30–31
 UV/chlorine vapor-phase cleaning for
 metal removal, 31

 removal of metal contaminants, 31
 Versatile process variables, 312
 Vertical brush scrubber configurations, 287,
 287f
 Vibration spectroscopies, 518
 “Vibrational state” of molecule. *See* Degree
 of vibration
 Volatile hydrides, 392–394
 Volatile organometallic complexes, formation
 of, 55
 Volatilization process, 314–315
 Volumetric
 instruments, 677
 particle counters, 687
 VPD. *See* Vapor phase decomposition (VPD)
 VPD/Atomic Absorption Spectroscopy
 (VPD/AAS), 706, 707t–709t, 714
 VPD/ICP–MS. *See* Vapor phase
 decomposition-inductively coupled
 plasma mass spectrometry
 (VPD/ICP–MS)
 VPD/TXRF. *See* Vapor Phase Decomposition
 Decomposition/Inductively Coupled
 Plasma Mass Spectroscopy
 (VPD/TXRF)

W

Wafer cleaning, 238–239
 alternative wet-chemical cleaning and
 surface conditioning systems, 21–25
 dry cleaning and surface conditioning
 processes, 28–36
 equipment for implementing wet-chemical
 cleaning, 26–27
 evolution of wafer cleaning science and
 technology, 36–60
 liquid processes and wafer drying
 techniques, 14–15
 period from (1950 to 1960), 37
 period from (1961 to 1971), 37–46
 radiochemical studies of surface
 contamination, 38–46, 38f
 period from (1972 to 1989)
 chronological survey of literature on
 H₂O₂-based cleans, 47–51
 other important advances, 51
 period from (1989 to 1992), 52–56
 vapor-phase cleaning methods, 54–56
 wet-chemical cleaning processes, 52–54
 period from (1993 to 2006)
 dry cleaning and surface conditioning
 processes, 59–60

Wafer cleaning (*Continued*)

- liquid processes and wafer drying
 - technology, 58–59
 - trends and milestones, 56–57
 - technology, 14–36
 - wafer rinsing, drying, and storing, 27–28
 - wet-chemical cleaning processes,
 - 15–21
- Wafer contamination, 88–101. *See also* Ultratrace impurity analysis of wafer surfaces
- aspects, 10–14
 - behavior and impact of contamination
 - AMC, 120–126
 - metallic contamination, 106–120
 - particle contamination, 101–106
 - classification and defects, 88–92
 - effects of contaminants and defectivity on
 - silicon device, 11–13
 - mechanisms of contamination, 96–101
 - basic aspects of wafer surfaces, 99–101, 99f
 - contamination transport through air, 97–98
 - contamination transport through liquids, 98
 - prevention of contamination and defectivity, 13–14
 - sources of defects and contamination, 126–141
 - types
 - and origins of contaminants and defectivity, 10–11
 - of semiconductor wafers, 11
 - yield models, reliability, and relationship to defectivity, 92–96
- Wafer(s), 187–188, 211, 340
- back surface defect inspection, 673
 - buff process, 294–295
 - carriers, 26–27
 - cassette, 229
 - cleaning and surface conditioning
 - for integrated circuit manufacturing, 7–9
 - technology, 6–7
 - defect and particle measurements on, 662–663
 - defect inspection equipment
 - patterned wafer defect inspection, 672–673
 - unpatterned wafer defect inspection, 669–672

- wafer back surface defect inspection, 673
 - wafer edge defect inspection, 674
 - drying, 27–28
 - edge defect inspection, 674
 - handling, transport, and carriers, 138
 - limits of analytical measurement
 - techniques, 674–677
 - particle detection, 695
 - particle monitoring on wafer for cleaning
 - tools, 664–668
 - particle scan maps with particle reattachment, 288, 288f
 - particulate contamination analysis on, 663–664
 - rinsing, 27–28
 - scanning process, 710–711
 - collection efficiency of scanning process, 711–712
 - storage, 28
 - surface, 660–661, 712
 - analysis, 705
 - basic aspects, 99–101, 99f
 - contamination on, 91–92
 - technology, 668–669
 - wafer-processing equipments, 430
 - wafer–slurry interface, 254–255
- Water (H₂O), 17, 318–320, 409
- H₂O-soluble form, 426
 - monitoring
 - and chemical, 620–621
 - conductivity and resistivity, 645–646
 - optical absorption spectroscopy, 626–627
 - rinsing fundamental aspects, 58–59
 - reaction with fluorine-terminated silicon surfaces, 568
 - vapor, 317
 - interaction with hydrogen-terminated silicon surfaces, 557–562
- “Watermarks”, 127–128, 225–227, 226f, 261–262
- Wavelength, 628–629
- emitted by UV sources, 340–342
 - ranges, 628–629, 630f
- Weibull distribution, 94
- Wet bench
- batch immersion processing tanks, 622–623
 - process, 666
- Wet cleaning
- chemical manufacturers, 193

- formulation, 412
 - limitations, 307–308
 - processes, 187, 653
 - Wet etching chemistries, 588–589
 - Wet processing, 620
 - chemicals, 624–626, 624t, 645
 - treatments, 606
 - Wet stations, particle monitoring of, 665–667, 667f
 - Wet wafer cleaning operations, 307
 - Wet-chemical, 5, 14
 - alternative wet-chemical cleaning and surface conditioning systems, 21–25
 - DDC, 24
 - IMEC Clean, 22–24, 24f
 - Ohmi Clean, 21–22, 22f
 - Single-Wafer/Short-Cycle Clean, 24–25, 25f
 - cleaning processes, 52–54, 416–417
 - alternative cleaning solutions, 19–21
 - equipment for wet-chemical cleaning implementation, 26–27
 - hydrofluoric acid solutions, 15–16
 - modifications of RCA cleaning process, 19
 - original RCA cleaning process, 17–18
 - sulfuric acid and hydrogen peroxide mixtures, 16–17
 - monitoring, 623–624
 - techniques, 522
 - Wet-cleaning, 29
 - Wettability tests, 338
- ## X
- X-ray fluorescence, 715
 - X-ray photoelectron spectroscopy (XPS), 329, 399, 414, 473–474, 473f, 508, 518, 520–522, 531–532, 586–587, 595, 704, 707t–709t, 730
 - spectra of polymer, 403, 404f
 - survey spectrum of Si surface, 331–333, 332f
 - Xenon (Xe), 463
- ## Y
- Yield
 - classification according to, 92
 - models relationship to defectivity, 92–96
 - defect size distribution, 95
 - effective area, 95–96
 - statistical yield models, 93–94
- ## Z
- Zeta potential, 163, 163f, 176–177, 202
 - Cu surface after cleaning with acidic and alkaline solutions, 282f
 - of materials as function of pH, 170f
 - of Si surface, 204
 - of wafer surfaces, 178f